IES ACADEMY • ENGINEERING

GenAI for Engineers

Version v2.4 · 19 October 2026

IES Academy

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COURSE ADMINISTRATION

Welcome & Housekeeping

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ICE-BREAKER

Let's Know Each Other

1. Your name, role and one maintenance reporting task.
2. Which app holds the work-order source?
3. Which date and status definitions determine overdue work?
4. Check which Copilot buttons and skills menus are available in your assigned apps.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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Access Course Materials

1. Open the IES Academy course folder supplied by your trainer
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the labs folder for all 12 lab folders
5. Keep the files together as you work

Course materials: trainer-supplied IES Academy folder

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SCHEDULE

Lesson Plan — 1 Day, 7.5 instructional hours

Course sequence

- Topic 1: Fundamentals of Microsoft 365 Copilot (Labs 1–2)
- Topic 2: Personalisation, .Rules and Skills for Excel (Labs 3–6)
- Topic 3: Interactive Dashboards with Custom Skills (Labs 7–8)
- Topic 4: Create a Professional Presentation with PowerPoint Copilot (Labs 9–10)
- Topic 5: Copilot Agents and Mini Capstone (Labs 11–12)

Day format

- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
  - 7.5 hours · 12 connected labs
- Learning arc
  - Concept → demonstrate → build → verify → discuss → recap

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Connected Copilot workflow

Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.

2

Reusable maintenance skills

Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.

3

Interactive dashboard

Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.

4

Professional presentation

Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.

5

Agents and capstone

Build and test a grounded work-order agent with Agent Builder, then apply the full data-to-decision workflow to your own mock or sanitised data in a group capstone.

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LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

TOPIC 1

Fundamentals of Microsoft 365 Copilot

One verified maintenance workflow across four apps

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Key Concepts — Fundamentals of Microsoft 365 Copilot

1

Large language models

Understand your request and generate natural-language answers.

2

Grounding

Answers anchored in your work data: emails, files, meetings.

3

Permissions & privacy

Shows only data you can access; prompts do not train the models.

4

Prompts

State the goal, context, source and expectations.

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WHAT YOU'LL DO

Hands-On Labs — Fundamentals of Microsoft 365 Copilot

Lab 1

- **Excel to Word**
  - Prepare Work Orders and Draft the Operations Brief

Lab 2

- **Word to PowerPoint and Outlook**
  - Create the Cross-App Maintenance Update

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CONNECTED USE CASE

The Northstar Maintenance Evidence Chain

1. Excel: clean and check
2. Word: operations brief
3. PowerPoint: decision story
4. Outlook: reviewed draft

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EDITABLE DATA MODEL

A Work Order Has State, Scope and Ownership

ID
WO-2026-0003

Site
Woodlands Hub

Category
Leak Repair

Status
In Progress

Owner
Chloe Goh

Current owner is evidence. Proposed owner is a separate planning output.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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The Microsoft 365 Copilot Interface

WHAT IT IS

- Copilot in the browser and in each Office app shares one set of controls.

WHY IT MATTERS

- Knowing where each control lives lets you choose how much reasoning, context and authority Copilot gets for a task.

CHECK IT

- Auto is the default model; Think deeper suits multi-step analysis.

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The Microsoft 365 Copilot Interface

1

Model selection

Auto, Quick response or Think deeper; Auto chooses for you.

2

New chat

Starts a clean conversation; each chat is its own context.

3

History

Earlier chats in the left pane to reopen or continue.

4

Mode

Excel/Word: Chat only, Plan or Allow editing (Edit).

5

+ Add content

Upload a local file, / a OneDrive file, or choose a skill.

6

Personalisation

Settings (…) > Personalization: your account preferences.

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The Microsoft 365 Copilot Interface (cont.)

7

Skills & plugins

Settings (…) > Manage plugins & skills; call one with @.

8

Agents

Agents & Skills in the left pane; build with Agent Builder.

9

Data protection

The green shield shows enterprise data protection is on.

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Two Ways to Give Copilot a File: Local or OneDrive

	Local file (this course)	OneDrive / SharePoint file
Where it lives	Your computer, e.g. the downloaded lab folder.	OneDrive or SharePoint in Microsoft 365.
In Excel	Desktop Excel Copilot works on the open workbook; no upload needed.	Excel desktop or web; AutoSave keeps the cloud copy current.
In Word, PowerPoint, Outlook	+ > Upload images and files, then pick the file on your device.	Type / or use + > Attach cloud files and pick the file.
/ or @ mention	Not available: / lists only cloud files; @ calls skills and agents.	/ lists the OneDrive and SharePoint files you can open.
After you edit	The upload is a copy: upload again after changing the file.	The reference reads the latest saved cloud version.
Use it for	Training labs and files you do not want in the cloud.	Shared team files, agent knowledge and folder references.

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The Microsoft 365 Copilot Interface

1. Pick the model and start a new chat.
2. Reference the file or choose a skill.
3. Choose Chat only, Plan or Allow editing.
4. Review history, settings and skills.

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Apply the Concept — The Microsoft 365 Copilot Interface

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Local file: upload it with +. OneDrive file: reference it with /.
- Chat only for inspection, Plan before edits, Allow editing to apply.

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Grounding Across Microsoft 365

WHAT IT IS

- Grounding gives an AI request the specific evidence it should use.

WHY IT MATTERS

- A maintenance figure can be correct in Excel yet become wrong in an email when the source, reporting date or filter scope changes.

CHECK IT

- Excel analyses tblWorkOrders; Word receives the checked summary.

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Grounding Across Microsoft 365

1. Identify the authoritative work-order table.
2. Provide goal, context, source and expected output.
3. Review against the named evidence.
4. Pass a verified snapshot to the next app.

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Apply the Concept — Grounding Across Microsoft 365

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- PowerPoint receives the reviewed Word brief; Outlook receives the same checked values.
- An Excel chat transcript is not an automatic PowerPoint data source.

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Work-Order Grain and Reporting Measures

WHAT IT IS

- A row represents one work order, identified by Work_Order_ID.

WHY IT MATTERS

- An exact duplicated export row changes counts and cost without representing new work.

CHECK IT

- 122 raw rows contain 120 distinct work-order IDs.

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Work-Order Grain and Reporting Measures

1. Define one ID per work order.
2. Find exact repeated export records.
3. Standardise labels without inventing facts.
4. Reconcile cleaned counts and cost.

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Apply the Concept — Work-Order Grain and Reporting Measures

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Total = Closed + Cancelled + Backlog after exact duplicate removal.
- Due on 30 September is not overdue at the 30 September snapshot.

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LAB 1 THEME · EXCEL TO WORD

Prepare Work Orders and Draft the Operations Brief

LAB 1

You are the maintenance coordinator at Northstar Water Services, a fictional water-pipe maintenance company. Clean a messy work-order export with Copilot in Excel, add check formulas, then turn the clean data into a one-page Word brief..

You'll build

A clean Northstar_Work_Orders.xlsx (120 rows and a Checks sheet) and Northstar_Operations_Brief.docx.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook

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Check Your Results — Lab 1

1

Clean table

120 rows, one per Work_Order_ID; 5 job categories.

2

Checks

120 total, 48 closed, 12 cancelled, 60 backlog, 50 overdue, 6 unassigned, cost S$35,827.50.

3

Word brief

Same numbers as the Checks sheet; empty owners left empty.

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Prompts and Verified App Handoffs

WHAT IT IS

- A useful prompt combines the intended goal, operational context, named source and output expectations.

WHY IT MATTERS

- The same number may have different meaning when a chart counts all work orders while an email calls it open backlog.

CHECK IT

- Goal: draft a maintenance update for the operations manager.

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Prompts and Verified App Handoffs

1. Write the goal and audience.
2. Upload the local file or paste the verified source.
3. Specify definitions and missing-value behaviour.
4. Reconcile the returned claims.

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Apply the Concept — Prompts and Verified App Handoffs

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Source: reviewed Northstar operations brief, all filters All, 30 September.
- Expectation: label assignment suggestions as proposals and leave unknown approvals UNKNOWN.

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Outlook Drafting and Operational Decisions

WHAT IT IS

- Copilot can help structure a draft email from supplied context.

WHY IT MATTERS

- An apparently routine update can accidentally promise technician reassignment or imply that a manager approved a recovery plan.

CHECK IT

- Current owner blank; proposed owner Aisha remains unapproved.

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Outlook Drafting and Operational Decisions

1. Provide the synthetic thread and checked facts.
2. Draft a concise update.
3. Review recipients, facts and action wording.
4. Save the unsent draft and evidence.

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Apply the Concept — Outlook Drafting and Operational Decisions

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A draft addressed to example.invalid is a training artifact.
- The handoff says decision requested, not decision approved.

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Create the Cross-App Maintenance Update

LAB 2

Turn the Northstar operations brief into a 4-slide PowerPoint update and an Outlook email draft, then check that all three show the same numbers..

You'll build

Northstar_Initial_Update.pptx (4 slides) and an unsent Outlook email draft.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook

© IES Academy

Check Your Results — Lab 2

1

Same numbers

Brief, slides and email all show 120 total, 60 backlog, 50 overdue and 6 unassigned.

2

Draft only

The email stays in Drafts; nothing is approved.

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TOPIC RECAP

Recap - Fundamentals of Microsoft 365 Copilot

1. LO1: Same work-order evidence across four apps.

2. Verify: 120 clean IDs and reconciled brief/update/draft.

TOPIC 2

Personalisation, .Rules and Skills for Excel

Account preferences, workbook conventions and reusable methods

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Key Concepts — Personalisation, .Rules and Skills for Excel

1

Personalisation

Your account preferences, applied across every workbook.

2

.Rules

Visible column-A instructions that travel with the workbook.

3

Skills

Reusable methods: built-in @chart-design, or your own SKILL.md such as Pareto and dashboard Skills.

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WHAT YOU'LL DO

Hands-On Labs — Personalisation, .Rules and Skills for Excel

Labs 3–4

- **Personalisation**
  - Personalise Copilot in Excel for Consistent Service Reports
- **Workbook .Rules**
  - Create and Test Workbook .Rules for Water Service Reporting

Labs 5–6

- **Custom Pareto Chart Skill**
  - Use the Built-in Chart Skill, then Create a Custom Pareto Skill
- **Reporting Skill**
  - Create, Use and Improve the Maintenance Reporting Skill

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EDITABLE SCOPE COMPARISON

Four Scopes of Customisation

1

Personalisation: your account

2

.Rules: this workbook

3

Skill: repeated method

4

Prompt: this request

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Account Personalisation and Specific Requests

WHAT IT IS

- Excel Copilot Personalisation stores account-level preferences for display, writing style and formula conventions.

WHY IT MATTERS

- Consistent S$ display and short explanations reduce repetitive requests without changing numeric values.

CHECK IT

- A value of 35827.5 can display as S$35,827.50 or S$35,827.5000.

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Account Personalisation and Specific Requests

1. Choose recurring display preferences.
2. Save them to the account.
3. Compare two workbook outputs.
4. Test a specific request and restore.

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Apply the Concept — Account Personalisation and Specific Requests

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Changing a number format does not change its numeric value.
- Another user opening the file does not receive your account preferences.

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LAB 3 THEME · PERSONALISATION

Personalise Copilot in Excel for Consistent Service Reports

LAB 3

Save personal preferences in Copilot, see Excel use them without being asked, change one for a single prompt, then put your settings back..

You'll build

A Preference_Check sheet that follows your saved preferences, and your original settings restored.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot in Excel

Check Your Results — Lab 3

1

Preferences used

Date 30-Sep-2026 and cost S$35,827.50 without asking in the prompt.

2

One-off change

S$35,827.5000 shown; saved preferences unchanged.

3

Restored

Your original Personalization settings are back.

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EDITABLE WORKBOOK MODEL

Visible .Rules: Instructions Travel with the Workbook

A1	Northstar reporting conventions
A2	Preserve original IDs and current owners
A3	Backlog excludes Closed and Cancelled
A4	Reporting date: 2026-09-30

Formula rule reads Control!B2; Copilot uses the recalculated result on the next request.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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Visible Workbook .Rules and Formula Results

WHAT IT IS

- A visible worksheet named exactly .Rules, including the dot, supplies workbook-specific Copilot instructions in column A.

WHY IT MATTERS

- The workbook can carry stable scope, status, cost-display and preservation instructions alongside the source.

CHECK IT

- The date rule returns Reporting date: 2026-09-30 from Control!B2.

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Visible Workbook .Rules and Formula Results

1. Name the visible sheet exactly .Rules.
2. Place one instruction per column-A cell.
3. Recalculate a formula-driven rule.
4. Check the next output and copied workbook.

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Apply the Concept — Visible Workbook .Rules and Formula Results

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Hidden .Rules is not used; keep the final sheet visible.
- No universal precedence among rules, preferences and Skills is assumed.

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Create and Test Workbook .Rules for Water Service Reporting

LAB 4

Give a workbook its own rules with a .Rules sheet, then see Copilot follow them without being told..

You'll build

A workbook with a visible .Rules sheet and a Rules_Test sheet that follows it.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot in Excel

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Check Your Results — Lab 4

1

.Rules sheet

Named exactly .Rules, visible, one rule per cell in column A.

2

Rules followed

Rules_Test: 120 total, 60 backlog, 50 overdue, 30/48 = 62.5%, S$35,827.50.

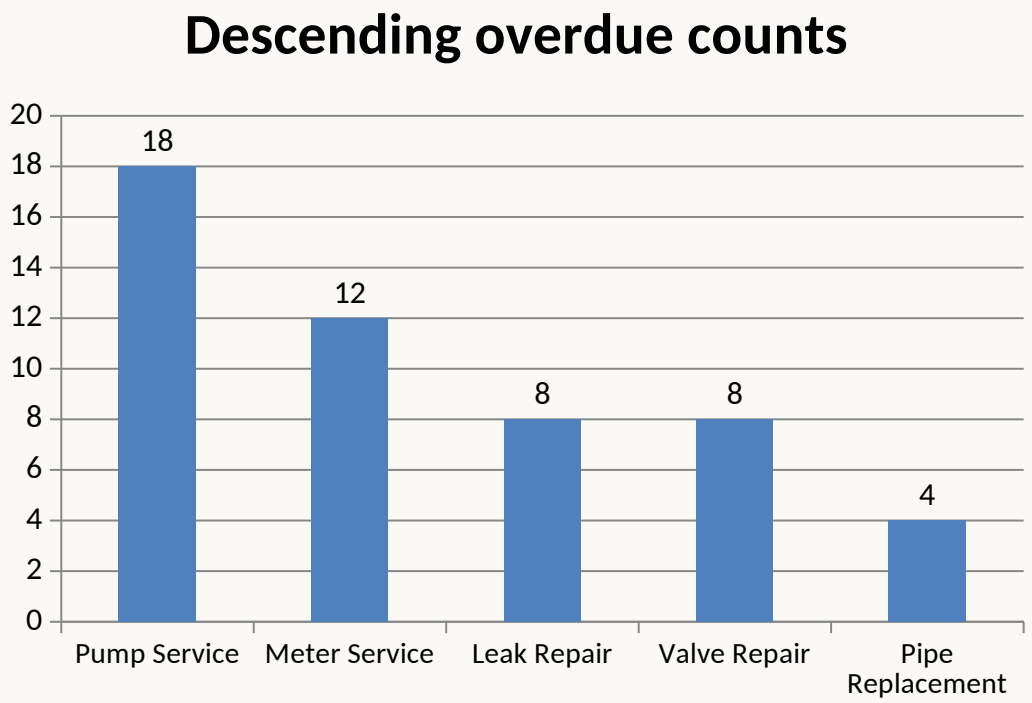
3

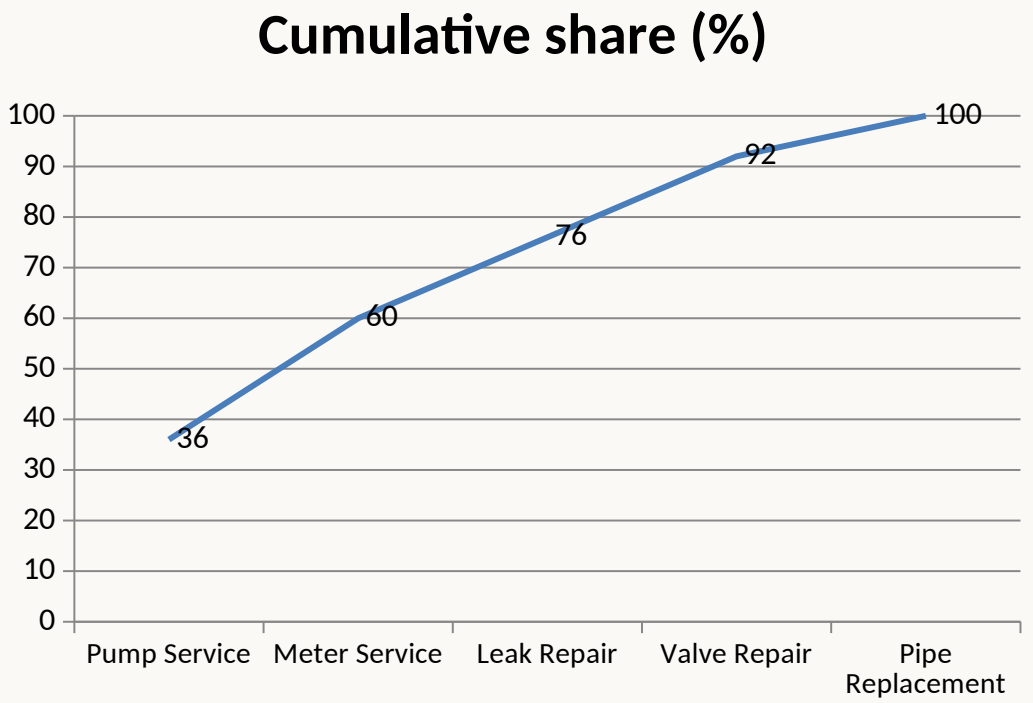
Copy

The saved copy still has the .Rules sheet.

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Overdue Contribution: 50 Work Orders, Five Categories





Four categories first reach 80%. Equal counts use alphabetical ties. Contribution is not cause.

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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Built-in and Custom Skills

WHAT IT IS

- Copilot in Excel ships with built-in skills you call with @, such as @chart-design for charts.

WHY IT MATTERS

- Start with a built-in skill when the task is generic, such as making a clear chart.

CHECK IT

- @chart-design draws a clean overdue-by-category bar chart.

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Built-in and Custom Skills

1. Turn on chart-design under Visualization.
2. Call @chart-design for a generic chart.
3. Spot the rules it cannot know.
4. Write a custom SKILL.md for those rules.

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Apply the Concept — Built-in and Custom Skills

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- It does not know the overdue population or Pareto tie order.
- water-service-pareto adds the cumulative line and named output.

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Pareto Skill: Counts, Sorting and Cumulative Share

WHAT IT IS

- A Pareto contribution chart orders category counts from largest to smallest and adds a cumulative-percentage line.

WHY IT MATTERS

- The Skill turns a recurring grouping-and-charting method into reusable instructions.

CHECK IT

- Pump 18; Meter 12; Leak 8; Valve 8; Pipe 4: total 50.

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Pareto Skill: Counts, Sorting and Cumulative Share

1. Define source, population, date and grouping.
2. Count and sort with stable ties.
3. Calculate cumulative count / eligible total.
4. Build and verify an editable dual-axis chart.

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Apply the Concept — Pareto Skill: Counts, Sorting and Cumulative Share

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Cumulative percentages are 36%, 60%, 76%, 92%, 100%.
- Four of five categories first reach 80%; do not force an 80/20 conclusion.

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Use the Built-in Chart Skill, then Create a Custom Pareto Skill

LAB 5

Make a chart with Excel’s built-in @chart-design skill, then install a custom Pareto skill and use it twice..

You'll build

A ChartDesign_Demo sheet, a Pareto_Job_Category chart and a Pareto_Site chart.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)

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Check Your Results — Lab 5

1

Built-in skill

chart-design turned on; ChartDesign_Demo created.

2

Pareto by category

50 jobs: 18, 12, 8, 8, 4; line 36%, 60%, 76%, 92%, 100%.

3

Pareto by site

Pareto_Site also adds up to 50; the source data is unchanged.

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EDITABLE SKILL CONTRACT

Anatomy of the Maintenance Skill

Discovery	name + description	When should Copilot use it?
Input guard	table or snapshot + date + scope	What evidence is missing?
Method	definitions + source preservation	How should measures be calculated?
Outputs	summary / charts / proposals	Which named areas may change?
Verification	counts + denominators + reruns	What proves the result?

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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REPEATABLE IMPROVEMENT

A Skill Improves Through Evidence

1. v1 instructions
2. Run on verified inputs
3. Find one concrete defect
4. v2 + repeat the check

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Skills, Prompts and Workbook Rules

WHAT IT IS

- A prompt is one request.

WHY IT MATTERS

- The skill makes stable definitions and review boundaries reusable: backlog excludes two terminal statuses, overdue uses a fixed date, missing owners stay missing and output reruns do not duplicate sheets.

CHECK IT

- maintenance-reporting contains definitions and output boundaries.

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Skills, Prompts and Workbook Rules

1. Identify a recurring task.
2. Separate stable rules from current inputs.
3. Write name, description and instructions.
4. Install, invoke and inspect the result.

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Apply the Concept — Skills, Prompts and Workbook Rules

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Each request still supplies this month’s source, date and scope.
- Native skill use and pasted manual instructions are recorded differently.

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Custom Skill Discovery in Each App

WHAT IT IS

- Excel uses its Manage plugins & skills interface and an app-created OneDrive skills folder.

WHY IT MATTERS

- Folder-name/frontmatter mismatches, an extra .txt extension, unavailable licence features and disabled skills can all prevent discovery.

CHECK IT

- Folder maintenance-reporting matches YAML name maintenance-reporting.

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Custom Skill Discovery in Each App

1. Save complete UTF-8 SKILL.md.
2. Use the app-created skills location.
3. Refresh and verify the matching name.
4. Run and record a smoke test.

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Apply the Concept — Custom Skill Discovery in Each App

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Excel and PowerPoint both show the skill before native invocation.
- If a menu is absent, manual practice is labelled feature unavailable.

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Repeatability and Safe Skill Reruns

WHAT IT IS

- Repeatability means the same verified inputs and method yield equivalent business measures and a controlled output structure.

WHY IT MATTERS

- Reporting skills may be used every week or month, so uncontrolled output accumulation quickly creates conflicting summaries.

CHECK IT

- 120 source IDs remain 120 after rerunning Maintenance_Summary.

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Repeatability and Safe Skill Reruns

1. Capture the source fingerprint and outputs.
2. Request a reviewed rerun plan.
3. Replace only named output areas.
4. Compare source and measures again.

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Apply the Concept — Repeatability and Safe Skill Reruns

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Missing report date causes a clarification request, not invented numbers.
- A second output sheet is a defect when replacement was requested.

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Assignment Proposals and Capacity Evidence

WHAT IT IS

- Assignment planning combines the work category, existing owner, estimated hours, technician primary category and available hours.

WHY IT MATTERS

- The same skill can improve Excel analysis and PowerPoint wording by preserving this boundary.

CHECK IT

- Unassigned work is counted only within nonterminal statuses.

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Assignment Proposals and Capacity Evidence

1. Identify unassigned backlog.
2. Compare category and available hours.
3. Record the proposal and assumptions.
4. Request the manager decision.

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Apply the Concept — Assignment Proposals and Capacity Evidence

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Available hours minus proposed effort is a planning check.
- The source Assigned_To stays unchanged until a real authorised operational update.

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LAB 6 THEME · REPORTING SKILL

Create, Use and Improve the Maintenance Reporting Skill

LAB 6

Install the maintenance-reporting skill and use it in Excel and PowerPoint..

You'll build

Maintenance_Summary and Assignment_Proposals sheets, and an improved Northstar_Initial_Update.pptx.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel and PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)

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Check Your Results — Lab 6

1

Summary

120 total; 60 backlog; 50 overdue; 30/48 = 62.5%; S$35,827.50.

2

Proposals

6 proposals; Assigned_To unchanged.

3

No data

The skill asks for data instead of making up numbers.

4

PowerPoint

4 slides, same numbers, clearer headlines.

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TOPIC RECAP

Recap - Personalisation, .Rules and Skills for Excel

1. LO2: Personalisation, .Rules, built-in @chart-design, Pareto and reporting Skills tested.

2. Verify: Preserved source plus missing-input and rerun tests.

TOPIC 3

Interactive Dashboards with Custom Skills

Create the Skill, build the dashboard and prove interactions

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Key Concepts — Interactive Dashboards with Custom Skills

1

Measures

Six formula KPIs with clear denominators and N/A rules.

2

Visual design

KPI tiles and charts whose titles state the finding.

3

Interactions

Dropdowns and slicers filter every tile and chart.

4

Refresh

Known deltas in an isolated test copy.

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WHAT YOU'LL DO

Hands-On Labs — Interactive Dashboards with Custom Skills

Lab 7

- **Custom Dashboard Skill**
  - Create and Apply a Custom Interactive Dashboard Skill

Lab 8

- **Interactive Filters and Refresh**
  - Test Dashboard Filters and a Data Refresh

WORKED DASHBOARD SNAPSHOT

Northstar Portfolio at a Glance

Total orders

120

Backlog

60

Overdue

50

Closed

48

On-time closed

30/48

Recorded cost

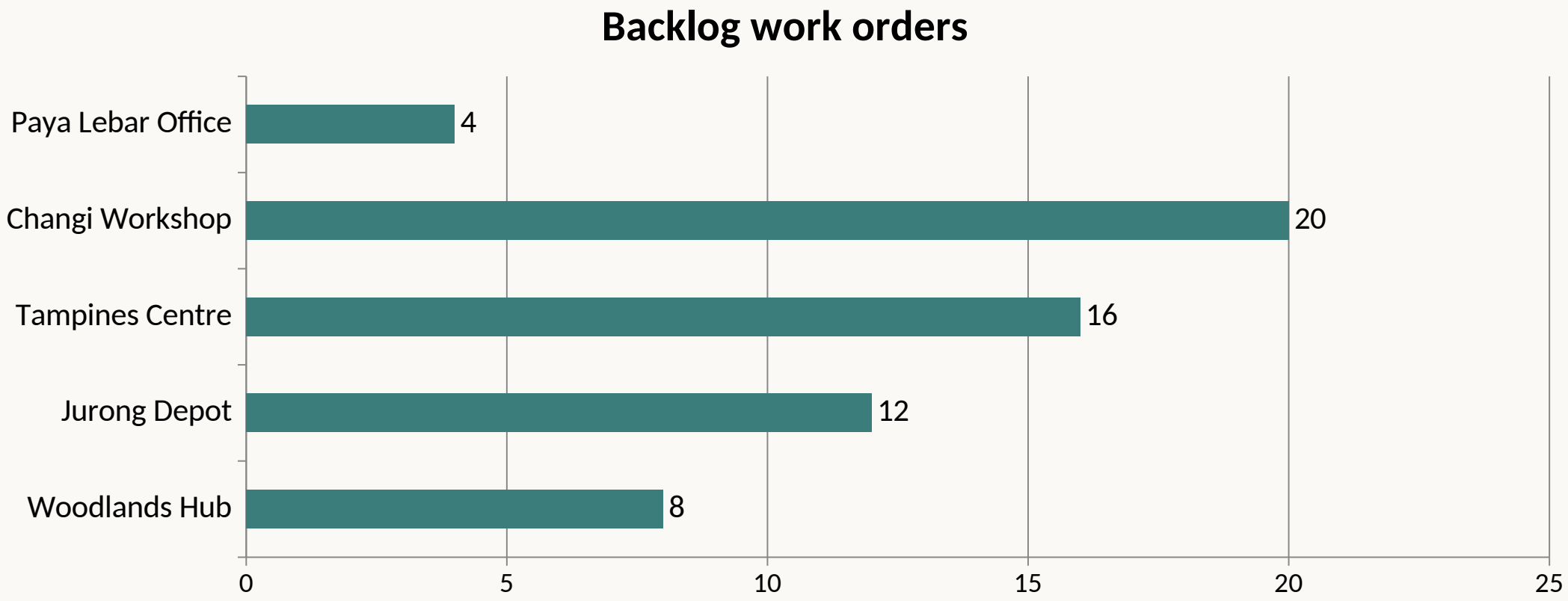
S$35,828

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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NATIVE EDITABLE CHART

Backlog by Site: Count the Nonterminal Work



Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

The Interactive Dashboard Skill Contract

1. Named source + date
2. Six shared-scope KPIs
3. Four filters + three charts
4. Refresh + safe rerun checks

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Custom Dashboard Skill as a Reusable Contract

WHAT IT IS

- water-service-dashboard is a separate custom Skill whose instructions define inputs, six KPI formulas, four shared filter selectors, three editable charts and controlled output boundaries.

WHY IT MATTERS

- A reusable contract makes every KPI and chart obey the same AND filter logic and allows a safe refresh using table references.

CHECK IT

- Dashboard B2:B5 select site, category, status and owner.

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Custom Dashboard Skill as a Reusable Contract

1. Author and review the complete SKILL.md.
2. Install and verify the dashboard name.
3. Invoke a plan and build the reviewed output.
4. Test common filters, append refresh and reruns.

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Apply the Concept — Custom Dashboard Skill as a Reusable Contract

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- B7:B12 and all chart summaries use the identical selected rows.
- A 122-row sandbox is separate from the untouched 120-row baseline.

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Dashboard Measures and Denominators

WHAT IT IS

- A dashboard measure needs a business definition, numerator, denominator where relevant, date rule and filter scope.

WHY IT MATTERS

- Changing the denominator can make an attractive rate meaningless.

CHECK IT

- Closed-only rate is different from closed/total completion share.

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Dashboard Measures and Denominators

1. Define source fields and scope.
2. Compute count and denominator.
3. Handle empty selections explicitly.
4. Reconcile against controls.

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Apply the Concept — Dashboard Measures and Denominators

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A due-today backlog item is not overdue under a strict before-date rule.
- All selected status/chart counts reconcile to total selected work orders.

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Visual Dashboard Design for Maintenance

WHAT IT IS

- A dashboard should make an operational decision easier.

WHY IT MATTERS

- Consistent chart scales, short labels, whitespace and limited accents help a manager compare the work queue quickly.

CHECK IT

- Status count chart shows the portfolio mix.

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Visual Dashboard Design for Maintenance

1. Name the maintenance decision.
2. Choose the measure and comparison.
3. Use an appropriate labelled chart.
4. Check meaning and readability.

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Apply the Concept — Visual Dashboard Design for Maintenance

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Site backlog chart excludes Closed and Cancelled.
- A category count does not establish the root cause of overdue jobs.

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Create and Apply a Custom Interactive Dashboard Skill

LAB 7

Install the water-service-dashboard skill and use it to build an interactive dashboard..

You'll build

Northstar_Dashboard.xlsx with six KPI tiles, three charts and slicers.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)

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Check Your Results — Lab 7

1

Tiles

120; 60; 50; 48; 62.5%; S$35,827.50.

2

Charts

Status, category and site charts each add up to 120; titles state a finding.

3

Formulas

No typed-in numbers; the source data is unchanged.

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EDITABLE WORKED COMPARISON

Same Data, Different Filter Scope

All sites

Total: 120

Backlog: 60

Overdue: 50

Closed: 48

On-time closed: 30/48

Woodlands Hub

Total: 24

Backlog: 8

Overdue: 7

Closed: 16

On-time closed: 10/16

Source: synthetic tblWorkOrders | 30 Sep 2026 | All filters All unless stated

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One Filter Scope for Every Visual

WHAT IT IS

- Interactive means a learner-controlled selection changes the evidence shown.

WHY IT MATTERS

- A dashboard can appear interactive while only one chart changes.

CHECK IT

- Woodlands Hub in B2 or in the Site slicer gives 24 work orders.

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One Filter Scope for Every Visual

1. Define the selector cells.
2. Apply the same AND mask everywhere.
3. Test each control and edge case.
4. Reconcile selected chart and KPI totals.

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Apply the Concept — One Filter Scope for Every Visual

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Status New gives zero Closed and an N/A closure rate.
- D3 warns while any slicer hides source rows.

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Refresh Tests and Static Snapshots

WHAT IT IS

- A refresh test deliberately introduces known new records and verifies the expected delta.

WHY IT MATTERS

- A correct dashboard today may fail after the next export if formulas reference a fixed cell range.

CHECK IT

- Two test rows increase total by2 and recorded cost byS$400.

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Refresh Tests and Static Snapshots

1. Save an isolated sandbox copy.
2. Append known unique test rows.
3. Recalculate and verify expected deltas.
4. Handoff the untouched baseline.

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Apply the Concept — Refresh Tests and Static Snapshots

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Only one new row adds to backlog/overdue; the other adds to on-time Closed.
- The final deck uses the untouched 120-row baseline, not the test copy.

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Test Dashboard Filters and a Data Refresh

LAB 8

Test a finished dashboard: filter it with slicers, check an empty result, then add two rows and watch every number update..

You'll build

Northstar_Refresh_Test.xlsx with the dashboard updated to 122 work orders.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Excel Copilot · Microsoft 365 Copilot web

© IES Academy

Check Your Results — Lab 8

1

Slicer

Woodlands Hub: 24 total, 8 backlog, 7 overdue, 16 closed, 10 of 16 on time, S$6,668.50.

2

Empty result

Status New: on-time rate N/A.

3

Refresh

122 total, 49 closed, 61 backlog, 51 overdue, S$36,227.50.

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Recap - Interactive Dashboards with Custom Skills

1. LO3: Six KPI tiles and three charts follow four dropdowns and slicers.

2. Verify: Baseline controls and known refresh deltas.

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TOPIC 4

Create a Professional Presentation with PowerPoint Copilot

Continue the dashboard into a decision pack

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Key Concepts — Create a Professional Presentation with PowerPoint Copilot

1

Evidence story

Observation, implication, action and decision.

2

Charts

Reviewed values, provenance and snapshot date.

3

Refinement

Hierarchy, notes and actual rehearsal feedback.

4

Handoff

Reconciled pack and unsent communication.

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WHAT YOU'LL DO

Hands-On Labs — Create a Professional Presentation with PowerPoint Copilot

Lab 9

- **Dashboard to presentation**
  - Turn Dashboard Evidence into a Management Story

Lab 10

- **Professional handoff**
  - Refine and Present the Maintenance Decision Deck

A Verified Chart Becomes a Decision

1. Observed backlog
2. Operational implication
3. Proposed recovery action
4. Manager decision requested

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From Dashboard Evidence to a Decision Story

WHAT IT IS

- A management presentation transforms evidence into a sequence of observations, implications, proposed actions and a decision request.

WHY IT MATTERS

- Copilot can accelerate outlines and drafting, but a persuasive invented conclusion is still wrong.

CHECK IT

- Backlog by site supports prioritisation, not a proven cause.

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From Dashboard Evidence to a Decision Story

1. Freeze the verified dashboard snapshot.
2. Register each proposed claim.
3. Write a reviewed Word brief.
4. Generate and reconcile the deck.

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Apply the Concept — From Dashboard Evidence to a Decision Story

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Available-hours evidence supports a capacity discussion, not confirmed allocation.
- The final slide asks management to decide on a proposed plan.

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Charts, Templates and Source Provenance

WHAT IT IS

- A template supplies slide layouts, typography and colours; it does not supply correct facts.

WHY IT MATTERS

- Approximate AI-generated charts can contradict source values while looking plausible.

CHECK IT

- An editable embedded chart remains a copy rather than a tested live link.

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Charts, Templates and Source Provenance

1. Upload or paste the reviewed brief.
2. Apply the existing template.
3. Insert verified charts or tables.
4. Record source and snapshot scope.

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Apply the Concept — Charts, Templates and Source Provenance

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- An image is labelled as a snapshot at 30 September.
- All eight slides use one reviewed baseline.

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Turn Dashboard Evidence into a Management Story

LAB 9

Turn the dashboard into a Word presentation brief, then into an 8-slide PowerPoint management deck..

You'll build

Northstar_Presentation_Brief.docx and an 8-slide Northstar_Management_Draft.pptx.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook

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Check Your Results — Lab 9

1

Brief

8 slide plans; 120 total, 60 backlog, 50 overdue and 6 unassigned.

2

Deck

8 slides; numbers match the brief; actions labelled proposed.

3

Chart

The dashboard site chart is on the site backlog slide.

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ARTIFACT LIFECYCLE

The Maintenance Reporting Handoff

1. Interactive workbook
2. Dated Word evidence
3. Static PPTX + PDF
4. Reviewed unsent email

Professional Refinement and Rehearsal

WHAT IT IS

- Refinement improves the clarity of the existing evidence without changing its meaning.

WHY IT MATTERS

- A well-designed deck can still fail when the speaker's notes contain different numbers or a proposal is spoken as an approved action.

CHECK IT

- One message per slide; chart labels remain readable.

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Professional Refinement and Rehearsal

1. Audit the draft against evidence.
2. Apply reviewed design changes.
3. Rehearse and record actual feedback.
4. Inspect PPTX and exported PDF.

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Apply the Concept — Professional Refinement and Rehearsal

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- A 45-second note explains evidence and limitation.
- Partner feedback leads to one recorded revision.

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Reviewed Handoff and Artifact Lifecycle

WHAT IT IS

- The final pack contains an interactive Excel dashboard, a fixed reporting brief, a static PowerPoint/PDF snapshot, the custom skill and a reviewed communication draft.

WHY IT MATTERS

- Updating one artifact without the others creates inconsistent communication.

CHECK IT

- PowerPoint and PDF both show eight slides/pages from the same baseline.

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Reviewed Handoff and Artifact Lifecycle

1. Reconcile final artifacts.
2. Record actual filenames and links.
3. Review the handoff draft.
4. Preserve the dated snapshot and skill.

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Apply the Concept — Reviewed Handoff and Artifact Lifecycle

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Outlook does not claim a missing attachment was sent.
- The next update reruns the method with new verified inputs.

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LAB 10 THEME · PROFESSIONAL HANDOFF

Refine and Present the Maintenance Decision Deck

LAB 10

Polish a management deck, rehearse it, save a PDF and draft the handoff email..

You'll build

Northstar_Management_Final.pptx and .pdf, and an unsent handoff email draft.

1 Ground	2 Build	3 Verify	4 Handoff

Tools: Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook

© IES Academy

Check Your Results — Lab 10

1

Deck

8 slides, same numbers, speaker notes added.

2

PDF

8 readable pages.

3

Email

Two attachments; stays in Drafts; nothing approved.

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TOPIC RECAP

Recap - Create a Professional Presentation with PowerPoint Copilot

1. LO4: Eight-slide evidence story and reviewed handoff.

2. Verify: PDF, notes, register and actual rehearsal agree.

TOPIC 5

Copilot Agents and Mini Capstone

Build a work-order agent, then apply the whole workflow in groups

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Key Concepts — Copilot Agents and Mini Capstone

1

Agent Builder

Create a specialist Copilot agent without code.

2

Knowledge

OneDrive files the agent searches and cites.

3

Instructions & Skills

Role, rules and a reusable work-order Q&A method.

4

Mini capstone

Groups turn safe data into a dashboard and decision deck.

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WHAT YOU'LL DO

Hands-On Labs — Copilot Agents and Mini Capstone

Lab 11

- **Copilot Agent Builder**
  - Build a Work-Order Agent with Agent Builder

Lab 12

- **Mini Capstone**
  - Mini Capstone: Group Dashboard and Decision Deck

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What Is Agent Builder?

WHAT IT IS

- Agent Builder is the no-code tool in Microsoft Copilot for creating your own specialist agent.

WHY IT MATTERS

- An agent packages a recurring job, here answering work-order questions, so colleagues get consistent, grounded and bounded answers without rewriting prompts.

CHECK IT

- Knowledge: the handbook and the clean work-order file from OneDrive.

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Agent Builder: What You Configure

1

Describe

Chat with Agent Builder to draft the agent in plain English.

2

Name & description

What the agent is for; shown to people you share it with.

3

Instructions

Role, sources, definitions, tone and boundaries (8,000 characters).

4

Knowledge

OneDrive/SharePoint files, uploads or websites it can search.

5

Skills

Reusable SKILL.md packages uploaded as a .zip (preview).

6

Try it & share

Test, select Create, then share as Can chat or Can edit.

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What Is Agent Builder?

1. Describe the agent’s job.
2. Configure instructions and knowledge.
3. Add a Skill and starter prompts.
4. Test, create and share safely.

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Apply the Concept — What Is Agent Builder?

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Skill: work-order-qa sets definitions, citations and UNKNOWN rules.
- Boundary test: the agent refuses to approve a reassignment.

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Build a Work-Order Agent with Agent Builder

LAB 11

Build a Copilot agent that answers questions about the Northstar work orders, then test its answers and its limits..

You'll build

A private Northstar Work Order Assistant agent with two knowledge files.

1 Describe	2 Configure	3 Ground	4 Test

Tools: Microsoft Copilot (copilot.cloud.microsoft) · Agent Builder · OneDrive

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Check Your Results — Lab 11

1

Answers

Backlog 60; overdue 50; a knowledge file is named.

2

Limits

Refuses to approve or change records; phone number UNKNOWN.

3

Private

The agent is not shared.

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Mini Capstone: From Your Data to a Decision

WHAT IT IS

- The mini capstone applies the whole course to a dataset your group chooses: safe data, workbook .Rules, a dashboard Skill, an interactive dashboard, a verified decision deck and a short showcase.

WHY IT MATTERS

- Repeating the workflow on unfamiliar data shows what transfers: definitions written as rules, reusable skills, independent checks and a decision-focused story.

CHECK IT

- A 150-row synthetic lift-maintenance table replaces real records.

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Mini Capstone: From Your Data to a Decision

1. Choose safe data and roles.
2. Write .Rules and build with a Skill.
3. Verify KPIs and slicers.
4. Build the deck and showcase.

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Apply the Concept — Mini Capstone: From Your Data to a Decision

PROMPT PRINCIPLES

- Name the source, reporting date and selected scope.
- Explain the result using checked evidence.

EVIDENCE TO CHECK

- Six KPI tiles reconcile to independent COUNTIFS.
- A five-slide deck ends with the decision requested.

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Mini Capstone: Group Dashboard and Decision Deck

LAB 12

In groups, build a dashboard and a 5-slide decision deck from your own fictional dataset, then present it in three minutes..

You'll build

Per group: a dataset, a .Rules sheet, an interactive dashboard, a 5-slide deck and a three-minute showcase.

1 Data	2 Rules	3 Dashboard	4 Deck	5 Showcase

Tools: Excel Copilot · PowerPoint Copilot · Microsoft 365 Copilot web

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Check Your Results — Lab 12

1. **Safe data**
   Fictional data only.

2. **Dashboard**
   Tiles and charts follow the slicers.

3. **Deck**
   Numbers match the dashboard; actions labelled proposed.

4. **Showcase**
   Three minutes, one live slicer click.

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TOPIC RECAP

Recap - Copilot Agents and Mini Capstone

1. LO5: A grounded work-order agent and a group dashboard and decision deck.

2. Verify: Agent answers match Control; capstone figures reconcile to the group data.

WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Connected Copilot workflow

Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.

2

Reusable maintenance skills

Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.

3

Interactive dashboard

Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.

4

Professional presentation

Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.

5

Agents and capstone

Build and test a grounded work-order agent with Agent Builder, then apply the full data-to-decision workflow to your own mock or sanitised data in a group capstone.

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NEXT STEPS

Your Next Maintenance Report

1. Use a copy and a new explicit reporting date.
2. Supply verified source/date/scope and test skill reruns.
3. Reconcile every claim and decision boundary.

Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded course files; every lab folder has its own starting workbook.
4. Keep practising with the IES Academy course materials.

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5 Principles of AI Security

1

Data control

Share only what the task needs; use synthetic or sanitised data.

2

Permission control

Least privilege for files, agents, skills and sharing.

3

Human in the loop

Plan, review and approve before edits, emails or decisions.

4

Traceability

Log prompts, sources, versions and checks for every output.

5

Well-tested models

Prefer proven, approved models over the newest release.

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THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.

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